Cloudra□□□□□□□□□□□□□□□□□□□□

– 2026 10 29

1. □□

1.1

exit 0

IRC

El Agente Q [1] Aitomia [2]

Gráfico [6]
Estructural [4] ARCHE [5]
Quntur [3]

Quntur [3]

——

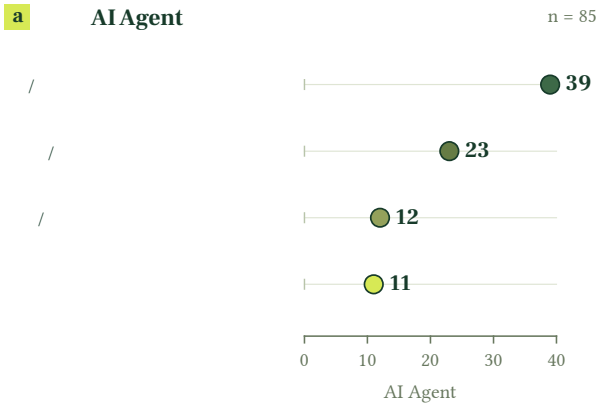
IRC

1.2 Cloudra

Cloudra—— Agent

Cloudra

1 Cloudra 85 AI Agent /
39 / 23 / 12 11 2
3 /



1: 85 AI Agent / 39 / 23 /
12 11

Cloudra “ ”

MCS

Cloudra Delivery

DAG Cloudra

IRC

Supervisor

d / n = 23 · 1/2

	AI Agent	T	E	O	F	R	D
01	Buehler molecular DFT	●	●	○	○	○	○
02	El Agente Q	●	●	●	●	○	○
03	Aitomia	●	●	●	●	○	○
04	MatAgent (Lv et al.)	●	●	○	○	○	○
05	ChemGraph	●	●	●	●	○	○
06	AutoSolvate-Agent	●	●	●	●	○	○
07	ChemAgents (azobenzene)	○	○	○	●	○	●
08	El Agente Cuántico	○	●	○	○	○	○
09	ChatMat	●	●	○	○	○	○
10	El Agente Estructural	●	○	●	○	○	○
11	El Agente Quntur	●	●	●	●	●	○
12	Chelatron	●	●	○	○	○	○
13	OpenClaw comp. chemistry	●	○	●	○	○	●
14	Protomia	●	●	○	●	○	○

● ○

2: / 1/2 23 AI Agent T /
E / O F / R / D /

d / n = 23 · 2/2

	AI Agent	T	E	O	F	R	D
15	El Agente Forjador	●	●	●	●	○	●
16	VirtualLab_CC	●	●	●	●	○	○
17	LADeQ	○	●	○	○	○	○
18	Evolving TS workflows	○	○	●	●	●	○
19	ARCHE	●	●	●	●	●	○
20	Multireference QC agent	○	●	○	○	○	○
21	SQD-Agent	○	●	○	○	○	○
22	QAOS / Quantum Agentic Operating System	●	●	●	○	○	○
23	PULSES	●	●	●	○	○	●

● ○

3: / AI Agent 2/2 2 85 AI
Agent

1.3

Cloudra Delivery DAG

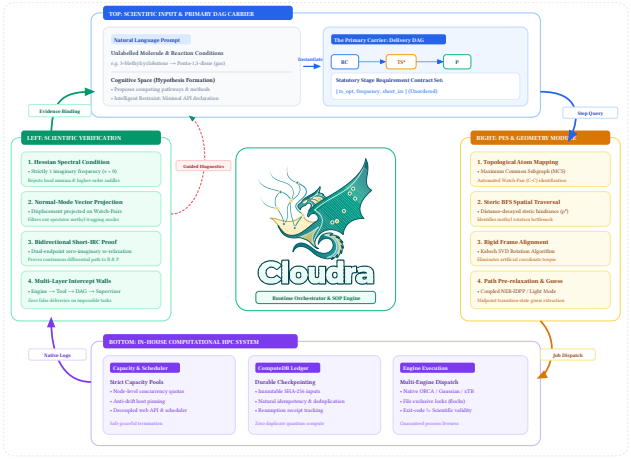
“ ”

CloudraBench Easy Normal Hard Impossible

2 □□□□□□□

2.1 □□□□□□□□□

Cloudra



4: Cloudra ComputeDB

2.1.1

`ComputeDB` SSH Slurm

SHA-256

`flock`

2.2 □□□□□□□□□

$H=(A,B)$ Cartesian $\mathbf{X}\in\mathbb{R}^{n\times 3}$ A B $\mathbf{X}$ H

2.2.1

ETKDGv3 [12,13]

A_P A_R $\pi: A_P \to A_R$ $z_i^P = z_{\pi(i)}^R$ $\iota_i^P = \iota_{\pi(i)}^R$

$$J(\pi) = 2(N_f + N_b) + N_o + 4\Delta_H + 8\Delta_q + 8\Delta_r + 2N_s. \quad (2.1)$$

B_P^π B_P^π B_R B_R B_P^π N_f N_b N_o $B_P^\pi \cap B_R$ $a \in \{H, q, r\}$ $\Delta_a = \sum_{i \in A_P} |a_i^P - a_{\pi(i)}^R|$ N_s 2

n_g $N_\Pi = \prod_g n_g!$ $N_\Pi \le 10^5$ MCS [13] MCS 10 64

16 C C $J_{min} = min_{\pi \in C} J(\pi)$ $\mathcal{N} = \{\pi \in C : J(\pi) \le J_{min} + 2\}$ $|\mathcal{N}| = 1$ $|\mathcal{N}| > 1$ $C = \varnothing$

h_g $N_H = \prod_g h_g!$ $N_H < 257$ $+2$

2.2.2

$\mathbf{x}_i$ $d_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$ j

$a - b$ b $\mathscr{F}_b$ $\mathscr{F}$ $\mathscr{F}_b$ $i \in \mathscr{F}$ Rodrigues

$\boldsymbol{x}_i' = \mathbf{a} + \mathbf{R}(\hat{\mathbf{u}}, \theta)(\boldsymbol{x}_i - \mathbf{a}), \mathbf{R} = \cos\theta\,\mathbf{I} + (1 - \cos\theta)$
$+ \sin\theta[\hat{\mathbf{u}}]_\times.$

(2.2)

$\boldsymbol{a}$ $\hat{\boldsymbol{u}}$ θ $\mathbf{I}$
$[\hat{\boldsymbol{u}}]_\times \mathbf{v} = \hat{\boldsymbol{u}} \times \mathbf{v}$
$a - b$ $\mathscr{F}_b$

$$\boldsymbol{x}_i' = \boldsymbol{x}_i + (L - d_{ab})\frac{\boldsymbol{x}_b - \boldsymbol{x}_a}{d_{ab}}, \quad i \in \mathscr{F}_b. \tag{2.3}$$

L

2.2.3

$$B_s = \sum_{k \in T_s} c_k r_k \rho^{\ell_k}, \quad \rho = 0.5, 0 \le \ell_k \le 5. \tag{2.4}$$

r_k k T_s
LIFO 0
ℓ_k $\ell_k < 5$
c_k 3 1.5 1.0
Å H 1.20, B 1.92, C 1.70, N 1.55, O 1.52, F 1.47, P 1.80, S 1.80, Cl 1.75, Br 1.85, I 1.98
1.70 B_s
$B_s \ge 9.0$

K s

$\hat{\boldsymbol{u}}_s = (a_s \cos s\gamma, y_s, a_s \sin s\gamma), a_s = \sqrt{1 - y_s^2}, y_s = 1 - \frac{2}{K}$

(2.5)

$\gamma = \pi(3 - \sqrt{5}) \approx 2.3999$ rad 137.5°
$\hat{\mathbf{u}}_s$ $2\pi s / K$ $K = 32$
W
$[l_{ij}, u_{ij}]$ $\boldsymbol{X}$

$$S(\boldsymbol{X}) = max_{(i,j) \in W} \{0, l_{ij} - d_{ij}(\boldsymbol{X}), d_{ij}(\boldsymbol{X}) - u_{ij}\}. \tag{2.6}$$

MMFF94

$d_{i^* j^*} \le 1.7(r_{i^*} + r_{j^*})$ 0.5 Å
Kabsch RMSD

2.2.4 **IDPP**

$\mathbf{P}$ $\boldsymbol{R}$

Kabsch

$min_{\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}, det\,\mathbf{Q} = 1} \|\mathbf{P}\mathbf{Q} - \boldsymbol{R}\|_F^2, \boldsymbol{P}^T \boldsymbol{R} = \mathbf{U}\,\Sigma\,\mathbf{V}^T, \boldsymbol{Q} = \mathbf{U}\,dia$

(2.7)

$\mathbf{Q}$
$det(\boldsymbol{U}\,\boldsymbol{V}^T)$ $det\,\boldsymbol{Q} = +1$
M $\lambda_k = k/(M-1)$
$\boldsymbol{X}_k = (1 - \lambda_k)\boldsymbol{R} + \lambda_k \boldsymbol{P}\boldsymbol{Q}$
IDPP [8]
$d_{ij,k}^* = (1 - \lambda_k) d_{ij}^R + \lambda_k d_{ij}^P$

$$d_{ij,k}^* = (1 - \lambda_k) d_{ij}^R + \lambda_k d_{ij}^P, V_k = \sum_{i<j} \frac{\left(d_{ij,k} - d_{ij,k}^*\right)^2}{d_{ij,k}^4}.$$

(2.8)

$d_{ij,k}^{-4}$

NEB
MDMin
$max(d_{ij,k}^*, 0.1\,\text{Å})^4$ Adam

W
$|\Delta d| \ge 0.15$ Å 5 W
$p^* \in argmax_{p \in W} |d_p^P - d_p^R|$
$m = (d_{p^*}^R + d_{p^*}^P)/2$ k
$(d_{p^*,k} - m)(d_{p^*}^R - m) \le 0$

2.3 □□□□□□□□□□□□

2.3.1

$G = (V, E)$ $v \in V$
Species Occurrence $e = (v_i, v_j) \in E$
$\kappa : V \to \mathcal{S}$

K
$K = \{\text{ts_opt}, \text{frequency}, \text{short_irc}\}$

$f = (c, o, r, d)$ c $o \in \{+, -, ?\}$
r d
R O t F_t

$S_t = 1[R \ne \varnothing] \wedge \bigwedge_{c \in R} 1[\exists f \in F_t : f.c = c, f.o = +] \wedge 1[\nexists f \in F_t :$
(2.9)

$R \ne \varnothing$ S_t $R = \varnothing$

2.3.2

Hessian

$$\mathbf{H}_{mw} = \mathbf{M}^{-1/2} \nabla_{\mathbf{x}}^2 U \mathbf{M}^{-1/2}, \mathbf{H}_{mw} \boldsymbol{v}_k = \lambda_k \boldsymbol{v}_k. \quad (2.10)$$

$\lambda_k < 0$ $v_k < 0$

$n_{imag} = \sum_k 1[v_k < 0]$ $n_{imag} = 0$

$n_{imag} = 1$

$\boldsymbol{X}$ $\mathbf{Y}$

$\delta_\pi(\boldsymbol{X}, \mathbf{Y}) = max_i \| \mathbf{Y}_{\pi(i)} \boldsymbol{Q}_\pi - \boldsymbol{X}_i \| \le 3 \times 10^{-6}$ Å

π Kabsch $\boldsymbol{Q}_\pi$

$\boldsymbol{u}$

W

$$\eta = \frac{max_{(i,j) \in W} \| \boldsymbol{u}_i - \boldsymbol{u}_j \|}{max_{i<j} \| \boldsymbol{u}_i - \boldsymbol{u}_j \|}. \quad (2.11)$$

$\eta \in [0,1]$ IRC

2.3.3 **IRC**

IRC $\mathbf{q} = \mathbf{M}^{1/2} \boldsymbol{x}$

[9,10]

$$\frac{d\mathbf{q}_\pm}{ds} = -\frac{\nabla_{\mathbf{q}} U}{\left\| \nabla_{\mathbf{q}} U \right\|}, \mathbf{q}_{init,\pm} \approx \mathbf{q}_{TS} \pm \epsilon \boldsymbol{v}_- . \quad (2.12)$$

IRC

$n_{imag} = 0$

SMILES

W IRC

$\boldsymbol{q}_{TS}$ $\boldsymbol{q}_E$ $\mathbf{q}_R$

$c_{ij} = 1.25(r_i + r_j)$ $\tau = 0.10$ Å

$W_c = \{(i,j) \in W : d_{ij}^R \le c_{ij}\}$

$[a_p, b_p] = [min(d_p^{TS}, d_p^R) - \tau, max(d_p^{TS}, d_p^R) + \tau]$

$is_between$ $\left\| \boldsymbol{q}_E - \boldsymbol{q}_R \right\|_2 < \left\| \boldsymbol{q}_{TS} - \boldsymbol{q}_R \right\|_2$

$a_p \le d_p^E \le b_p$ $p \in W_c$

$is_between$

2.3.4

e K_e $P_{e,k} = 1$

k 0

$$A_e = \bigwedge_{k \in K_e} P_{e,k}. \quad (2.13)$$

R $A_e \wedge S_e$

$$\Delta E_e^{\ddagger} = E_{TS,e} - E_{ref,e}. \quad (2.14)$$

Gibbs

2.4 □□□□□□□□□□

SCF

$n_{imag} \neq 1$ $\eta < \eta_{thresh}$ IRC

T $P_{e,k}$ $t \in T$

$$U = \{(t,k) : t \in T, k \in K_t, P_{t,k} = 0\}. \quad (2.15)$$

U U

SCF IRC

U

2.5 □□□□□□

Cloudra 70daa5a

32 8 1.0 Å

1.7 RMSD 0.5 Å MMFF94

2000 100.0 IDPP 10

MDMin 100 $f_{max} = 0.1$ NEB

$k = 0.1$ IDPP Adam 4000 0.02

10^{-7}

3 □□□□□

Cloudra

DAG

3.1

CloudraBench 50 Easy Normal Hard Impossible 20 10 10 10

Diels–Alder

48 Guided Record Standard Brief-errors

E1 NB1 3- HA1 IA1

6,087

DeepSeek-v4-flash Qwen3.8:27b

token

3.2

3.2.1

SMILES 16

64

58 32 4 32 6 A 5 C_5H_8

A A “ ” “ ”

Cloudra

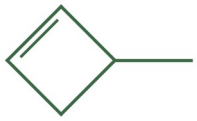
a Chemical identity across notation and option order

Pair / task system	Source strings Fwd	Source strings Swap	Equivalent rewrite Fwd	Equivalent rewrite Swap
Same identity				
01 E1 · reactants	A	B	A	B
02 E1 · products	A	B	A	B
03 NB1 · reactant	A	B	A	B
04 NB1 · products	A	B	A	B
05 HA1 · cyclic reactant	A	B	A	B
06 HA1 · E declaration	A	B	A	A
07 HA1 · Z declaration	A	B	A	A
08 IA1 · reactant	A	B	A	B
Same formula · changed connectivity				
10 NB1 · reactant / products	B	A	B	A
11 HA1 · cyclic / acyclic	B	A	A	A
13 NB1 · connectivity control	B	A	B	A
14 HA1 · connectivity control	A	A	A	A
15 IA1 · ring / chain	B	A	B	A
Connectivity + charge relocation				
09 E1 · reactants / products	B	A	B	A
Specified alkene stereochemistry				
12 HA1 · E / Z	B	A	A	A
Protonation + formal charge				
16 NB1 · protonation control	B	A	B	A

A Correct identity A Wrong identity Identical strings

b Same formula, different identity

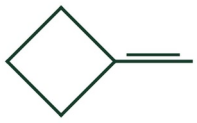
Pair 14 · HA1 reactant



CC1CC=C1

C_5H_8

Connectivity control



C=C1CCC1

C_5H_8

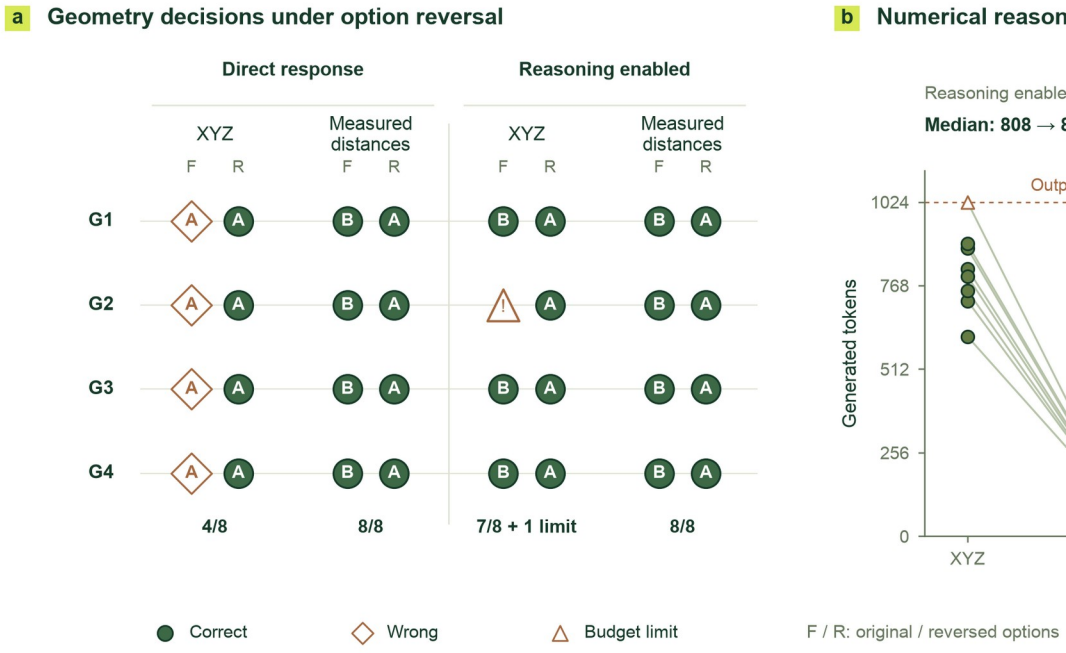
Option order	A	B
Forward	Same	Different
Swapped	Different	Same

5. 16 14 C_5H_8

3.2.2

NB1 XYZ
32 6
XYZ A XYZ 1,024
token 1,353 token XYZ token
807.5 86

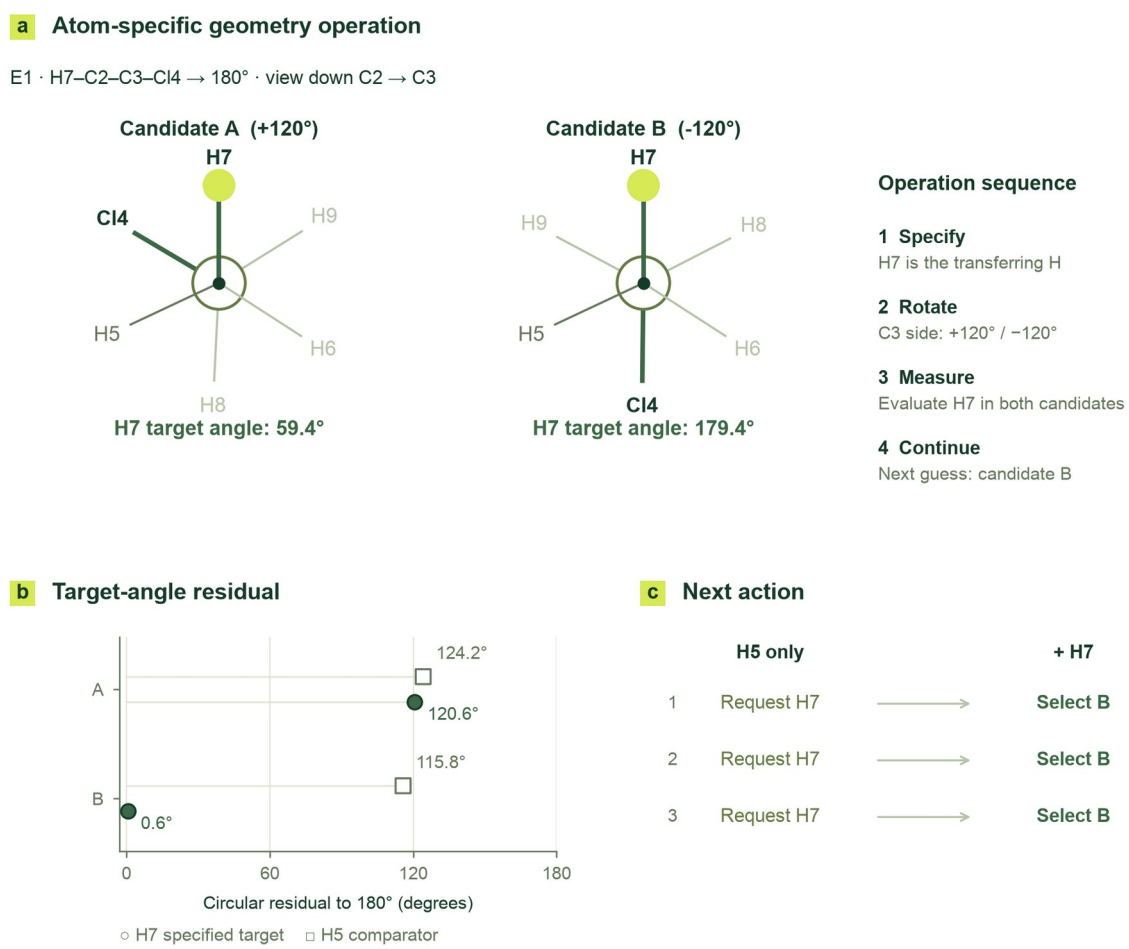
Cloudra
[15]



6. token

3.2.3

E1 H7 C2 C3 Cl4 180° C2–C3 C3
+120° −120° A B C3–Cl4 2.35 Å H5 H7 H7–C2–C3–
Cl4 59.4° 179.4° B
7 C2→C3 Newman B H7 180° 0.6° A 120.6° H5
124.2° 115.8°
H5 H7 H7 B



7. E1 A B Newman H7 H7 H5 B

3.2.4

[16] 2.029 Å 1.029 Å 1.248 Å

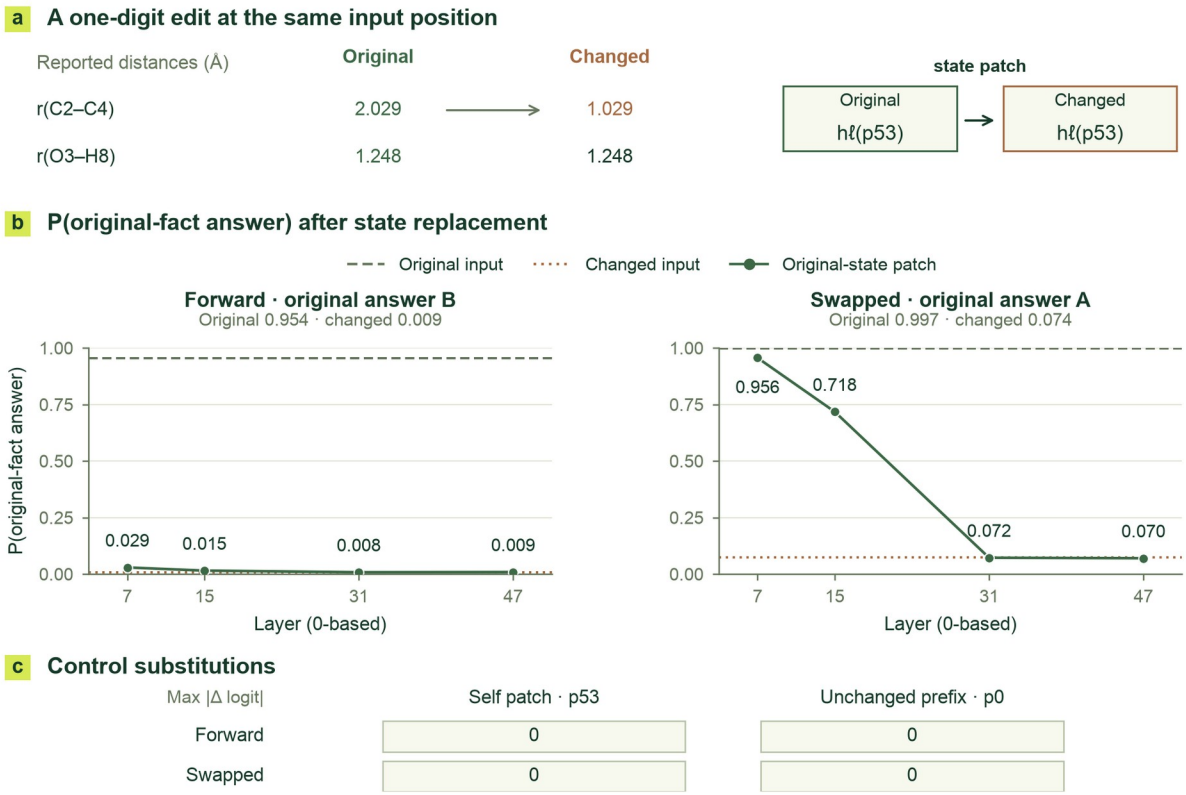
token 7 15 31 47

8 A 0.074 7 15 0.956 0.718 31 47

0.072 0.070 B 0.009 0.029 0.015 logit

“ ” “ ”

Cloudra



8. token

3.3

3.3.1 DAG

IRC

DAG

32 IRC

9a E1

16 16 14 E1 NB1 16

[17] Cloudra

3.3.2

NB1 2,500

Brief-errors HA1 IRC Guided

NB1

“ ” “ ” [18]

Cloudra

3.3.3

IA1 DAG Supervisor

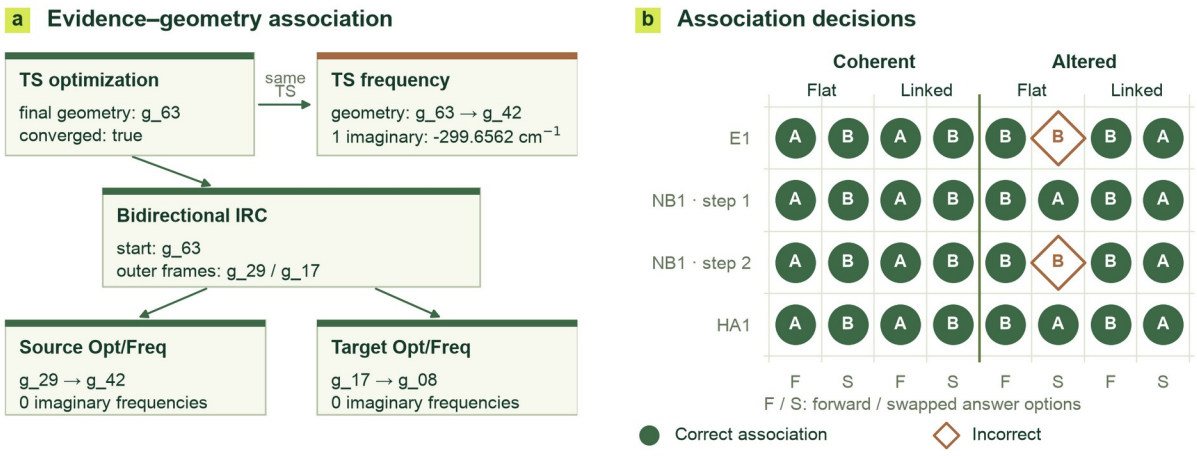
short_irc 9c

12 12 12

DAG Supervisor

DAG

Cloudra



9. a E1 b 32 c IA1 DAG

□□□□

[1] Zou, Y., et al. El Agente: An autonomous agent for quantum chemistry. Matter 2025, 8, 102263. https://doi.org/10.1016/j.matt.2025.102263

[2] Hu, J., et al. Aitomia: An Agentic Framework for AI-Driven Atomistic and Quantum Chemical Simulations. Journal of Chemical Theory and Computation 2026, 22, 6557–6571. https://doi.org/10.1021/acs.jctc.6c00591

[3] Pérez-Sánchez, J. B., et al. El Agente Quntur: A research collaborator agent for quantum chemistry. Chem 2026, 103259. https://doi.org/10.1016/j.chempr.2026.103259

[4] Choi, C., et al. El Agente Estructural: An Artificially Intelligent Molecular Editor. 2026. https://arxiv.org/abs/2602.04849

[5] Li, D., et al. Autonomous Chemical Mechanistic Discovery through Agentic Reasoning and Validation. 2026. https://arxiv.org/abs/2609.11147

[6] Bai, J., et al. El Agente Gráfico: A Semantic Execution Runtime for Scientific Agents. arXiv:2602.17902v2, 2026. https://arxiv.org/abs/2602.17902v2

[7] Kabsch, W. A discussion of the solution for the best rotation to relate two sets of vectors. Acta Crystallographica Section A 1978, 34, 827–828. https://doi.org/10.1107/S0567739478001680

[8] Smidstrup, S.; Pedersen, A.; Stokbro, K.; Jónsson, H. Improved initial guess for minimum energy path calculations. The Journal of Chemical Physics 2014, 140, 214106. https://doi.org/10.1063/1.4878664

[9] Fukui, K. Formulation of the reaction coordinate. The Journal of Physical Chemistry 1970, 74, 4161–4163. https://doi.org/10.1021/j100717a029

[10] Maeda, S.; Harabuchi, Y.; Ono, Y.; Taketsugu, T.; Morokuma, K. Intrinsic reaction coordinate: Calculation, bifurcation, and automated search. International Journal of Quantum Chemistry 2015, 115, 258–269. https://doi.org/10.1002/qua.24757

[11] FACCTs. ORCA 6.1 Manual: Vibrational frequencies and Intrinsic reaction coordinate. https://www.faccts.de/docs/orca/6.1/manual/contents/structurereactivity/frequencies.html

[12] Riniker, S.; Landrum, G. A. Better Informed Distance Geometry: Using What We Know To Improve Conformation Generation. Journal of Chemical Information and Modeling 2015, 55, 2562–2574. https://doi.org/10.1021/acs.jcim.5b00654

[13] RDKit. Getting Started in Python: Conformer Generation and Maximum Common Substructure. https://www.rdkit.org/docs/GettingStartedInPython.html

[14] Cloudra . 70daa5a. https://github.com/yyywaa/Cloudra-mechanism-agent/tree/70daa5a

[15] Gao, L.; Madaan, A.; Zhou, S.; et al. PAL: Program-aided Language Models. Proceedings of the 40th International Conference on Machine Learning 2023, 202, 10764–10799. https://proceedings.mlr.press/v202/gao23f.html

[16] Zhang, F.; Nanda, N. Towards Best Practices of Activation Patching in Language Models: Metrics and Methods. International Conference on Learning Representations 2024. https://proceedings.iclr.cc/paper_files/paper/2024/hash/06a52a54c8ee03cd86771136bc91eb1f-Abstract-Conference.html

[17] Fatemi, B.; Perozzi, B.; Halcrow, J. Talk like a Graph: Encoding Graphs for Large Language Models. International Conference on Learning Representations 2024. https://research.google/pubs/talk-like-a-graph-encoding-graphs-for-large-language-models/

[18] Gou, Z.; Shao, Z.; Gong, Y.; et al. CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing. International Conference on Learning Representations 2024. https://arxiv.org/abs/2305.11738